

# M E R N Technology Stack

shopEZ — B2C E-Commerce Marketplace (MERN Stack)

|                 |                                                 |
|-----------------|-------------------------------------------------|
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Technology Stack

# 1. Introduction

This document provides a formal specification of the technology stack selected for the ShopEZ platform. Each technology selection is accompanied by its architectural role, justification, and a brief Architecture Decision Record (ADR) summarising the decision rationale and alternatives considered.

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## 2. Integration Architecture Diagram

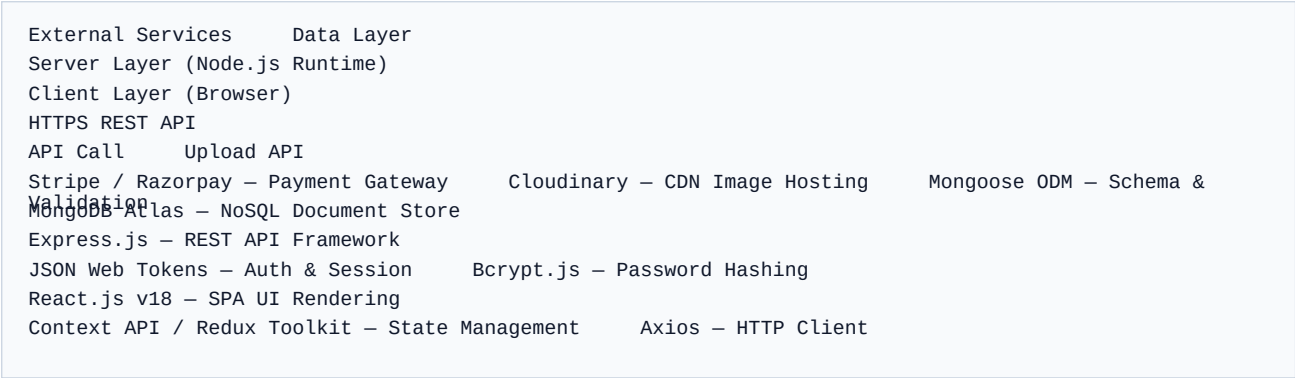


FIGURE 5.1 — TECHNOLOGY STACK INTEGRATION ARCHITECTURE

DIAGRAM

## 3. Detailed Stack Breakdown

| Layer              | Technology                     | Version | Purpose & Justification                                                                                                            |
|--------------------|--------------------------------|---------|------------------------------------------------------------------------------------------------------------------------------------|
| Frontend Framework | React.js                       | v18.x   | Component-driven SPA architecture. Virtual DOM ensures efficient UI re-rendering. Hooks-based API enables clean functional design. |
| State Management   | Redux Toolkit / Context API    | v2.x    | Centralised global state for user authentication sessions, shopping cart, and cached API data. Redux Toolkit reduces boilerplate.  |
| HTTP Client        | Axios                          | v1.x    | Promise-based HTTP client with request/response interceptors, automatic JSON serialisation, and base URL setup.                    |
| Backend Runtime    | Node.js                        | v18 LTS | JavaScript runtime enabling full-stack single-language development. Non-blocking, event-driven I/O suits concurrent REST handling. |
| Backend Framework  | Express.js                     | v4.x    | Minimal, unopinionated framework providing middleware composition, route parameters, and error handlers for REST APIs.             |
| Authentication     | JSON Web Tokens (JWT) RFC 7519 |         | Stateless, self-contained bearer tokens signed with server secret. Stored in HTTP-only cookies to mitigate XSS and CSRF.           |
| Password Security  | Bcrypt.js                      | v2.x    | Industry-standard adaptive hashing (salt rounds >= 10) applied via pre-save hooks on the User model.                               |
| Database           | MongoDB (Atlas)                | v6.x    | Document-oriented NoSQL store for flexible schema design. Hosted on Atlas for managed auto-scaling and backups.                    |
| ODM                | Mongoose                       | v8.x    | Schema-based ODM providing validation, type enforcement, virtual fields, populate references, and lifecycle hooks.                 |
| Payment Gateway    | Stripe / Razorpay              | Latest  | PCI-DSS compliant payment APIs. No sensitive financial data stored on the local database.                                          |
| Media Hosting      | Cloudinary SDK                 | v2.x    | Provides automatic image compression, WebP conversion, and CDN delivery. Eliminates local filesystem dependency.                   |
| Deployment         | Vercel (FE) / Render (BE) -    |         | Vercel provides React static hosting with global CDN. Render hosts Node.js API with automated git-based deployments.               |

## 4. Architecture Decision Records (ADRs)

### ADR-01: MongoDB over Relational Database

| Field         | Detail                                                                                                                                          |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| Status        | Accepted                                                                                                                                        |
| Context       | The product catalogue requires a flexible schema - product attributes vary significantly across categories (electronics, apparel, home living). |
| Decision      | Adopt MongoDB as the primary data store.                                                                                                        |
| Justification | Document-based storage natively accommodates variable product attributes without migration-heavy ALTER TABLE operations.                        |
| Consequences  | No multi-document ACID transactions by default. Mitigated by Mongoose document-level atomicity for order creation and stock deduction.          |

















